Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–June 29th, 2026

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

A. What can be said about the time complexity of a PRAM-based application when the PRAM model has a bounded number of processors? (1)

B. Please define what is meant by "Work" in a PRAM application. (1)

C. How do the different types of concurrent write (CW) models affect algorithm design in PRAM? (1)

D. Please describe how Amdahl and Gustafson define the speed-up. (1)

Exercise n. 2

A. Please describe the key properties of the Karger & Stein algorithm. How is the complexity analyzed? (2)

B. Please discuss the complexity of the "Powering a number" algorithm. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. Please list the main differences between the Las Vegas and Monte Carlo algorithms. What about the efficiency? (1)

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Exercise n. 3

A. Please describe how the treap union works. (1)

B. What are the differences between the randomized and deterministic algorithms for solving the Select-ith problem (pseudo code required)? (1)

C. Please describe the difference between online and offline algorithms. How is their complexity analyzed? (2)

Exercise n. 4

A. Please explain the differences between the *computed table* and the *unique table* in the context of BDDs. Which one is more closely related to dynamic programming algorithms? (2)

B. Could you explain which aspects of primality testing use randomization and how that process works? Additionally, would you classify it as a Las Vegas randomized algorithm? (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**